09_Rank of a Matrix in Linear Algebra

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The rank of a matrix is the maximum number of linearly independent rows or columns in the matrix. It represents the dimension of the column space or row space of the matrix. Let's see the column rank method:

$$\text{For Example: } A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

Step 1: Identify the Column Vectors

$$C1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \qquad C2 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \qquad C3 = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

We check if these columns are linearly dependent by expressing one column as a combination of others.

- Observe that C2 = 2C1 and C3 = 3C1.
- Since all columns are multiples of C1, there is only one linearly independent

column.

Thus, the **rank of A is 1.**

Properties of rank

- A square matrix A of size n x n is full rank if its rank is n (i.e., all rows/columns are linearly independent).
- If a matrix has dependent rows or columns, its rank is less than the total number of rows or columns.
- If A is a square matrix and det(A) $\neq 0$, then A has full rank n.
- If det(A) = 0, the matrix is singular and has rank < n.
- If det(A) = 0, the matrix has dependent rows or columns, meaning its rank is less than n (not necessarily 0). Det(A) = 0 does NOT mean rank is 0
- A matrix has rank = 0 only if all its elements are zero.

$$A = \begin{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{bmatrix}$$

- Here, all rows and columns are dependent (zero vectors), so rank = 0.

Alternative to Matrix Division

If A is a **square, invertible matrix**, we define matrix "division" as:

$$A^{-1} \times A = I$$

So instead of $A \div B$, we write:

$$X = A^{-1}B \quad \Bigg| \quad \begin{array}{l} ax = b \quad \Rightarrow \quad x = \dfrac{b}{a} = a^{-1}b \text{ (Scaler)} \\ AX = B \end{array}$$

where A^{-1} is the **inverse** of matrix A.

The Inverse Matrix Acts Like an Alternative to Division

Inverse Of a Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \Bigg| \quad \det(A) = ad - bc \quad \Bigg| \quad A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

(Formula)

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\det(A) = (2 \times 4) - (3 \times 1) = 8 - 3 = 5$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

(Inverse is Done)

Finally, $X = A^{-1}B$

$$X = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} \longrightarrow X = \begin{bmatrix} \frac{4}{5} \times 5 + (-\frac{3}{5} \times 6) \\ -\frac{1}{5} \times 5 + \frac{2}{5} \times 6 \end{bmatrix}$$

$$X = \begin{bmatrix} \frac{20}{5} - \frac{18}{5} \\ -\frac{5}{5} + \frac{12}{5} \end{bmatrix} \longrightarrow X = \begin{bmatrix} 0.4 \\ 1.4 \end{bmatrix}$$

(Final Result)

$$A = \begin{bmatrix} 3 & 5 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 2 \\ 7 & 5 \end{bmatrix}$$

Comment below the Trace of $A^{-1}B$

$$A^{-1}B = \begin{bmatrix} -11/7 & -17/7 \\ 15/7 & 13/7 \end{bmatrix}$$

$$\mathrm{Trace}(A^{-1}B) = \left(-\frac{11}{7}\right) + \left(\frac{13}{7}\right)$$

$$= \frac{-11+13}{7} = \frac{2}{7} \approx 0.2857$$